EPA 608 Universal Certification — Personal Study Guide

Prepared for: William Marceau **Compiled from:** ESCO 9th Edition practice exam misses (Type I, Type II, Type III) + session study notes **Target:** Pass EPA 608 Universal certification (Core + Type I + Type II + Type III in one sitting)

How to use this guide

Three parts:

1. **Top priority study areas** — the topics where you missed the most questions. Drill these first.
2. **Complete reference tables** — every memorization-heavy fact, in table form
3. **Concept explanations** — the "why" behind the numbers for everything we discussed

Read sections 1-2 cold the night before the exam. Section 3 is for understanding during prep, not last-minute review.

CRITICAL UPDATE — You can bring the ESCO PT chart to the exam

The two-page ESCO Pressure-Temperature chart is permitted in the exam room. That chart contains an enormous amount of information you would otherwise have to memorize. Knowing exactly what is and isn't on the chart **changes your study strategy completely**.

What's ON the chart (do NOT waste time memorizing these — just look them up)

Information on the chart	Skip memorizing
ASHRAE safety class of every common refrigerant (A1, A2L, A3, B1, B2, etc.)	Class column headers above each refrigerant
GWP value of every common refrigerant	GWP row above each refrigerant
Pressure category (low/medium/high/very-high)	Determined by which chart section the refrigerant appears in, color-coded

Refrigerant family (CFC, HCFC, HFC, HFO, Hydrocarbon, etc.)	Listed under each refrigerant column
Pressure-temperature relationships	The grid of P/T values at every 5° increment
Whether a refrigerant is a blend with glide	Blends show "Bubble Liquid" and "Dew Vapor" columns

The ESCO chart's pressure category boundaries

The chart divides refrigerants by **pressure at 104°F condensing temperature**:

Section	Pressure at 104°F
Low-pressure	< 30 psig
Medium-pressure	30 - 155 psig
High-pressure	155 - 340 psig
Very-high-pressure	> 340 psig

When a question asks "what pressure category is R-X?" — just find R-X on the chart, see which colored section it's in. Done.

The refrigerants explicitly on your chart

Refrigerant	Chart shows	Chart shows	Chart shows
R-22	High-pressure	A1	GWP 1810
R-404A	High-pressure	A1	GWP 3920
R-407C	High-pressure	A1	GWP 1770
R-422D, R-422B	High-pressure	A1	GWP 2730 / 2530
R-441A	High-pressure	A3	GWP <5 (hydrocarbon)
R-410A	High-pressure	A1	GWP 2090
R-717 (ammonia)	High-pressure	**B2**	GWP 0
R-744 (CO_2)	**Very high-pressure**	A1	GWP 1 (base)
R-123	Low-pressure	**B1**	GWP 77
R-245fa	Low-pressure	B1	GWP 1030
R-1233zd	Low-pressure	A2L	GWP 4.7-7

R-12	Medium-pressure	A1	GWP 10,900
R-124	Medium-pressure	A1	GWP 609
R-409A	Medium-pressure	A1	GWP 1558
R-134a	Medium-pressure	A1	GWP 1430
R-600a (isobutane)	Medium-pressure	A3	GWP 3
R-1234yf	Medium-pressure	A2L	GWP 4
R-1234ze	Medium-pressure	A2L	GWP 6

Important observation: R-134a appears in the **medium-pressure** section on this chart (its 104°F pressure is in the 30-155 range), even though it's sometimes loosely grouped with "high-pressure HFCs." **Trust the chart** for classification questions on the exam.

What's NOT on the chart (THIS is what you must memorize)

The chart doesn't give you any of the EPA regulatory information. That's where ALL your remaining study time should go:

Memorize	Because the chart doesn't show it
EPA recovery vacuum levels by pressure category + charge size	Chart shows category but not the EPA recovery requirement
Type I recovery percentages (90% / 80%) and vacuum (4" Hg)	No EPA recovery thresholds on the chart
EPA recordkeeping triggers (5 lbs, 15 lbs, 50 lbs, 200 lbs)	Not on chart
Leak repair timelines (30 days, 10 days, 12 months, 18 months)	Not on chart
Leak rate thresholds (20%/year comfort cooling, 30%/year commercial)	Not on chart
Low-pressure system specifics (10 psig leak test max, 15 psig rupture disc, 25 mm Hg absolute)	Not on chart
Recovery procedures (system-dependent vs self-contained, liquid-first then vapor, compressor on/off, high-side vs low-side connections)	Not on chart
Charging locations (low-pressure systems charge at evaporator low point)	Not on chart
Component functions (compressor, condenser, metering device, evaporator)	Not on chart

Saturation, superheat, subcooling concepts	Chart shows P/T values but not how to interpret them as superheat/subcooling
Ozone depletion mechanism (UV, C-Cl bond, catalytic destruction)	Not on chart
ASHRAE classification meanings (what A1, A2L, B1 actually mean)	Chart shows the labels but not what they mean for service decisions
Refrigerant family rules (Rule of 90, decoding R-numbers)	Useful if a question mentions a refrigerant not on the chart
EPA certification scope (Type I/II/III/Universal)	Not on chart
Phaseout timeline (CFC 1996, HCFC 2020, HFC 2024-2036)	Not on chart
HC refrigerant applications and charge limits (R-600a in household fridges, 57g limit)	Not on chart
Compressor burn-out diagnostic	Not on chart
Non-condensables vs moisture	Not on chart

[redacted]

Given the chart is in your exam toolbox:

Time allocation	Topic
60%	EPA regulatory facts (vacuum levels, timelines, thresholds, recordkeeping) — Priority 1, 2, 4 in Part 1
20%	Procedures (recovery, charging, leak detection, burn-out diagnostic) — Priority 3 in Part 1
10%	Conceptual understanding (cycle components, saturation, ODP/GWP mechanism) — Part 3
5%	Rule of 90 + family identification (for refrigerants not on chart)
5%	Familiarize yourself with the chart layout so you can find things FAST during the exam

Chart-reading practice (do this before exam day)

1. **Find R-123 on the chart.** Note: low-pressure section, B1 classification, GWP 77.
2. **Find R-744 (CO_2).** Note: very-high-pressure section, A1, GWP 1.
3. **Find R-1234yf.** Note: medium-pressure section, A2L, GWP 4 (HFO).
4. **Find R-410A's pressure at 104°F.** Should be 336 psig (right at the high-pressure boundary).

5. **Find R-600a (isobutane).** Note: medium-pressure section, A3, GWP 3 (hydrocarbon).

Speed matters on the real exam — knowing the chart layout cold means you spend 5 seconds looking something up instead of 30 seconds.

How this changes your previous misses

If you'd had the chart available during the practice exams, you could have correctly answered:

Previously missed question	Chart provides the answer
Q11 Type II: ASHRAE class of R-12, R-22, R-134a	All three labeled **A1** on the chart
Q15 Type III: Which refrigerant is B1	Chart shows **R-123 = B1** explicitly
Q21 Type III: R-134a ASHRAE classification	Chart shows **A1**
Q25 Type II: Very high-pressure refrigerant	Chart's far-right column shows **R-744 / CO_2** as very-high-pressure
Q22 Type II: R-407C recovery vacuum at 210 lbs	Chart confirms R-407C is **high-pressure** → apply 10" Hg rule (which you must memorize)

Net effect: Of the 20 questions you missed across all three practice exams, **5 were chart-lookup questions**. With the chart available, you would have only missed 15 — an improvement from 73% to 80% [redacted]

If you also memorize the EPA regulatory facts in Part 1 + Part 2 of this guide, you should easily score 85-90%+ on the real exam.

PART 1 — TOP PRIORITY STUDY AREAS (based on your misses)

Across three practice exams (75 questions total), your wrong answers clustered in **five specific areas**. Drill these before anything else.

Priority 1: Recovery vacuum levels (you missed 5+ questions on this)

This is the #1 area to drill. The EPA has different recovery vacuum requirements based on:

- The pressure category of the refrigerant
- The charge size
- The age of the recovery equipment

You consistently picked "15 something" as a default answer. STOP that. 15 is rarely the right answer outside of medium-pressure ≥200 lbs (which is rare in modern HVAC).

The complete EPA recovery vacuum table (memorize cold)

For equipment manufactured **AFTER November 15, 1993**:

Pressure category	< 200 lbs charge	≥ 200 lbs charge
Very high-pressure (R-13, R-23, R-503, R-744 / CO_2)	0 psig	0 psig
High-pressure (R-22, R-32, R-134a, R-404A, R-407C, R-410A, R-507A)	**0" Hg vacuum**	**10" Hg vacuum**
Medium-pressure (R-12, R-114, R-500)	10" Hg vacuum	15" Hg vacuum
Low-pressure (R-11, R-123)	**25 mm Hg absolute**	**25 mm Hg absolute**
Type I — Small appliance (≤5 lbs)	**4" Hg vacuum or 90% recovery** (operating compressor) / **4" Hg vacuum or 80% recovery** (non-operating compressor)	(Type I systems are always ≤5 lbs by definition)

Memory shortcut

> ***"0 / 10 / 15 / 25" going down the categories (post-1993)***
>
> - *0 psig for very high-pressure (any size)*
> - *0 → 10" Hg for high-pressure (small → big at 200 lb threshold)*
> - *10 → 15" Hg for medium-pressure (small → big)*
> - *25 mm Hg absolute for low-pressure (any size — "absolute" not gauge)*

Refrigerants you need to instantly classify by pressure

Refrigerant	Pressure category	Where you'll see it
R-11	Low-pressure	Old chillers (CFC, phased out)
R-123	Low-pressure	Modern chillers (HCFC, phasing out)
R-12	Medium-pressure	Old auto AC, refrigerators (CFC, phased out)
R-114, R-500	Medium-pressure	Rare older systems
R-22	High-pressure	Old residential AC (HCFC, phased out for new equipment)
R-134a	High-pressure	Auto AC, some chillers

R-410A	High-pressure	Most modern residential AC
R-32, R-404A, R-407C, R-507A	High-pressure	Various commercial/residential
R-13, R-23, R-503	Very high-pressure	Cascade/ultra-low temperature systems
R-744 (CO_2)	Very high-pressure	Emerging modern (transcritical systems)

Priority 2: Low-pressure system specifics (Type III)

You missed multiple Type III questions involving low-pressure system numbers. Here are the four numbers you MUST commit to memory:

Number	What it is	When to use it
10 psig	**Maximum leak test pressure** for low-pressure systems	"When pressurizing a low-pressure system for leak detection..."
15 psig	**Rupture disc burst pressure**	"At what pressure does the rupture disc rupture?" — DO NOT confuse with leak test max
25 mm Hg absolute	**Required recovery vacuum** for low-pressure	"Recover to a level of..."
50 ppm	**OEL for R-123** (B1 classification)	Why R-123 is B-class toxicity

Drill the rule: "Pressurize to 10. Disc bursts at 15. Recover to 25 mm absolute. R-123 OEL is 50 ppm."

Low-pressure system specifics summary

Property	Value/Detail
Refrigerants in this category	R-11 (CFC, banned), R-123 (HCFC, phasing out), R-1233zd (HFO replacement)
Normal operating evaporator pressure	Below atmospheric (vacuum)
Normal operating condenser pressure	Slightly above atmospheric (~15-20 psia)
Why deep recovery vacuum needed	Refrigerant boils near room temp at atmospheric pressure
Charging location	**Lowest access point on the evaporator** (evaporator charging valve, bottom of evaporator)

Recovery sequence	**Liquid first, then switch to vapor**
Recovery driving force	External recovery machine — NOT the chiller's compressor
Burn-out indicator	Strong odor during recovery (acids + burned oil)
Required ventilation	Enhanced for R-123 (B1 class) — refrigerant monitor required
Purge unit purpose	Remove non-condensables (air) that leaked in (since system runs in vacuum)

Priority 3: Self-contained vs system-dependent recovery (Type I)

You missed multiple questions on this terminology. Get this once and for all.

Term	Other name	How it works	When used
Self-contained	"Active" recovery	**Has its own internal pump/ compressor.** Pulls refrigerant out independently of the system.	Any system — required when the system compressor is broken
System-dependent	"Passive" recovery	**No internal pump.** Uses the system's working compressor to push refrigerant into the recovery container.	ONLY on appliances with **15 lbs of refrigerant or less** AND only when the system compressor is operating

Procedure for system-dependent recovery with operating compressor

1. Install access fitting on the **high side** (discharge line)
2. Connect recovery container/cylinder
3. **Run the system's compressor** — it pumps refrigerant FROM the system INTO the cylinder

Procedure for system-dependent recovery with non-operating compressor

1. Install access fittings on **BOTH high and low sides**
2. Use **pressure differential, gravity, sometimes heat on the system / cold on the cylinder** to drive transfer
3. Slower, more steps, more setup

Priority 4: EPA recordkeeping + leak repair timelines

These are pure-memorization regulatory facts that appeared on Type II and Type III:

Threshold/ Timeline	What it means
≤ 5 lbs	"Small appliance" definition (Type I scope)
≥ 5 lbs	Recordkeeping starts being required
≤ 15 lbs	System-dependent recovery is legal at this charge or below
≥ 50 lbs	Leak repair rule activates (must repair if leak rate exceeds threshold)
≥ 200 lbs	Stricter recovery vacuum requirements (10" Hg high-pressure / 15" Hg medium-pressure)
30 days	Deadline to repair leak OR develop retrofit/retire plan after exceeding leak rate
10 days	Initial verification test after leak repair
30 days	Follow-up verification test after leak repair
12 months	Standard retrofit/retire completion timeline
18 months	Extended retrofit/retire timeline when replacement uses ≥50 lbs of **exempt** refrigerant
20%/year	Leak rate trigger for comfort cooling (HVAC)
30%/year	Leak rate trigger for commercial refrigeration and industrial process refrigeration
Monthly records	EPA requires monthly (not annual) records for 5+ lb appliances

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Priority 5: Pressure category identification

This was the underlying cause of most recovery vacuum misses. **You must know which refrigerants are** [redacted]

Drill these flashcards:

High-pressure (most common — modern HVAC)

- R-22, R-32, R-134a, R-404A, R-407C, R-410A, R-507A, R-502
- HCFCs and HFCs typically
- Operating pressures: ~50-400 psig

Medium-pressure (mostly legacy)

- R-12, R-114, R-500
- Old CFC refrigerants
- Rare in modern field work

[redacted]

- R-11, R-123, R-1233zd, R-113
- Centrifugal chillers, large commercial HVAC
- Boiling point > 50°F at atmospheric

[redacted]

- R-13, R-23, R-503, R-744 (CO_2)
- [redacted]
- Operating pressures > 1000 psig possible

PART 2 — COMPLETE REFERENCE TABLES

[redacted]

ASHRAE Standard 34 — Safety Classifications

Toxicity classification (the letter)

Letter	Toxicity	OEL threshold
A	Lower toxicity	≥ 400 ppm
B	Higher toxicity	< 400 ppm

Flammability classification (the number)

Number	Flammability	Criteria
1	No flame propagation	Won't burn in air
2L	Lower flammability ("slightly flammable")	LFL > 0.10 kg/m³ AND burning velocity ≤ 10 cm/s
2	Lower flammability	LFL > 0.10 kg/m³ but burning velocity > 10 cm/s
3	Higher flammability	LFL ≤ 0.10 kg/m³ OR high heat of combustion

Common refrigerants by ASHRAE class

Class	Meaning	Common refrigerants
A1	Non-toxic, non-flammable	R-11, R-12, R-22, R-134a, R-404A, R-407C, R-410A, R-507A, R-744 (CO_2), R-1233zd

A2L	Non-toxic, slightly flammable	**R-32, R-1234yf, R-1234ze, R-454B, R-452B**
A2	Non-toxic, lower flammability	R-152a
A3	Non-toxic, highly flammable	**R-290 (propane), R-600a (isobutane), R-1270 (propylene)**
B1	Toxic, non-flammable	**R-123** (50 ppm OEL), R-245fa
B2L	Toxic, slightly flammable	**R-717 (ammonia)**

Refrigerant Family Identification

How to identify each family

Family	Composition	Identification
CFC (Chlorofluorocarbon)	C + Cl + F (no H)	R-11, R-12, R-13, R-113, R-114
HCFC (Hydrochlorofluorocarbon)	C + H + Cl + F	R-22, R-123, R-124, R-141b, R-142b
HFC (Hydrofluorocarbon)	C + H + F (no Cl)	R-32, R-125, R-134a, R-143a, R-152a, R-404A, R-407C, R-410A, R-507A
HFO (Hydrofluoroolefin)	C + H + F (no Cl) with C=C double bond	R-1234yf, R-1234ze, R-1233zd, R-1336mzz
Hydrocarbon (HC)	C + H only	R-290 (propane), R-600a (isobutane), R-1270 (propylene), R-50 (methane)
Natural inorganic	No carbon	R-717 (ammonia), R-718 (water), R-744 (CO_2)

R-number range decoder

Range	Meaning
R-10 to R-29	Methane-based (1 carbon)
R-100 to R-199	Ethane-based (2 carbons)
R-200 to R-299	Propane-based (3 carbons)
R-400 to R-499	**Zeotropic blends** (have temperature glide)

R-500 to R-599	**Azeotropic blends** (behave like single refrigerant)
R-600 to R-699	Organic compounds (typically hydrocarbons)
R-700 to R-799	Inorganic naturals (R-717 ammonia, R-718 water, R-744 CO_2)
R-1000+	**Olefins (HFOs)** — have C=C double bond

Rule of 90 (decode molecular formula)

> ***R-number + 90 = (C)(H)(F)*** *as a 3-digit code.*
>
> *Remaining bonds filled by chlorine.*
>
> *Bonding sites = 2·C + 2 (so 4 sites for methane-based, 6 for ethane-based)*

Examples:

[redacted]

Environmental Impact — ODP & GWP

[redacted]

Ozone Depletion Potential (ODP) — relative to R-11 = 1.0

Family	ODP	Why
CFC (R-11, R-12)	**1.0** (the reference)	Multiple Cl + no H + long atmospheric life
HCFC (R-22, R-123)	0.02 - 0.11	Has Cl, but H makes it less persistent (reacts with OH in troposphere)
HFC (R-134a, R-410A, R-32)	**0**	No chlorine at all
HFO (R-1234yf, R-1233zd)	**0**	No chlorine, plus C=C double bond makes it very reactive
Naturals (CO_2, NH_3, hydrocarbons)	**0**	No halogens

Global Warming Potential (GWP) — relative to CO_2 = 1

Refrigerant	Family	GWP
R-744 (CO_2)	Natural	**1** (reference)
R-1234yf	HFO	< 1

R-1233zd	HFO	≈ 1
R-290 (propane)	HC	3
R-600a (isobutane)	HC	3
R-32	HFC	675
R-134a	HFC	1,430
R-22	HCFC	1,810
R-410A	HFC blend	2,088
R-23	HFC	14,800 (very high)

Mechanism summary

[redacted]

- [redacted] → free Cl destroys ozone catalytically)
- **High GWP = C-F bonds (strong IR absorbers) + long atmospheric lifetime** — independent of chlorine
- **HFOs win on both axes** because (a) no Cl = no ODP, (b) C=C double bond = short atmospheric lifetime = low GWP despite C-F bonds

[redacted]

Family	Phased out under	Year
CFC	Montreal Protocol	1996 production banned
HCFC	Montreal Protocol (amendment)	2020 (US new production)
HFC	AIM Act (Kigali Amendment)	2024-2036 phasedown (85% reduction)
HFO	Current/future	Not phased out

EPA Recovery Equipment Standards

Type I (Small Appliance ≤ 5 lbs)

Compressor status	Recovery percentage	OR achievable vacuum
Operating	**90%**	**4" Hg vacuum**
Non-operating	**80%**	**4" Hg vacuum**

All recovery equipment manufactured **after November 15, 1993** must meet these standards and be EPA-certified.

Type II (High-Pressure & Medium-Pressure)

See Priority 1 table above.

Type III (Low-Pressure)

All low-pressure systems: **25 mm Hg absolute** (regardless of charge size).

[redacted]

System type	Charging location
Low-pressure (R-11, R-123 chillers)	Lowest access point on the **evaporator** (evaporator charging valve, bottom of shell)
High-pressure (most residential AC)	Suction service port
Type I small appliances	Process tube or schrader valve, typically on the suction line

[redacted]

Recovery Procedures Summary

[redacted]

1. System compressor OFF (isolated)
2. Connect external recovery machine to liquid charging valve (evaporator bottom)
3. **Liquid recovery first** (machine in liquid mode) — pulls bulk of charge
4. Once liquid flow stops → **switch to vapor recovery mode**

[redacted]

6. Document recovery (type, weight, date)

Type I small appliance recovery (system-dependent, operating compressor)

[redacted]

2. Connect recovery container
3. **Run system compressor** — pumps refrigerant into container
4. Continue until 90% recovered or 4" Hg vacuum

[redacted]

Type I small appliance recovery (system-dependent, non-operating compressor)

1. Install access fittings on **BOTH high and low sides**
2. Use pressure differential, gravity, sometimes heat/cold

3. Recover until 80% or 4" Hg vacuum

Type II high-pressure recovery

1. Use self-contained (active) recovery machine
2. Connect to liquid line and/or suction line
3. ~~Recover until required vacuum reached (0" Hg or 10" Hg per the table)~~

Leak Repair Rule Summary

[redacted]

System type	Trigger
Comfort cooling (HVAC)	**20%/year**
Commercial refrigeration	**30%/year**
Industrial process refrigeration	**30%/year**
Other (50+ lbs)	**30%/year**

[redacted]

Action	Deadline
Notify EPA (if industrial process refrigeration)	30 days
Repair the leak OR develop retrofit/retire plan	30 days
Initial verification test of repair	10 days
Follow-up verification test	30 days
Complete retrofit/retire (standard)	**12 months**
Complete retrofit/retire (≥50 lbs exempt refrigerant)	**18 months**

[redacted]

For appliances **5+ lbs of refrigerant**, owners must keep **monthly records** of:

- Refrigerant added (by type)

[redacted]

- Refrigerant sent for reclamation or destruction (by type)

Non-Condensables vs. Moisture

Contaminant	What it is	Symptoms	Removal
Non-condensables	**Air** (N_2 + O_2)	Higher head pressure than expected at ambient temp	Deep vacuum (evacuation)
Moisture	Water vapor	Ice plugs at metering device, acid formation, oil breakdown	Filter-drier (during operation) + deep vacuum (during service)

Important: "Non-condensable" in HVAC terminology specifically means **air**, not water. Water DOES condense at refrigeration temperatures; air doesn't.

[redacted]

Component	What enters	What leaves	What it does
Compressor	Low-pressure vapor (cool, slightly superheated)	High-pressure vapor (hot, highly superheated)	Raises pressure. No phase change.
Condenser	High-pressure superheated vapor	High-pressure subcooled liquid	Rejects heat. Vapor → liquid phase change.
Metering device (TXV, piston, EEV)	High-pressure subcooled liquid	Low-pressure liquid+vapor mix	Drops pressure. Some refrigerant flashes to vapor.
Evaporator	Low-pressure liquid+vapor mix	Low-pressure superheated vapor	Absorbs heat. Liquid → vapor phase change.

Saturation, Superheat, and Subcooling

Definitions

Term	What it means	Where it exists
Saturation temperature	Temperature at which liquid ↔ vapor phase change happens at a given pressure	Only during phase change (middle of condenser and evaporator)
Superheat	Temperature above saturation, in vapor form	Compressor inlet (~8-20°F) and outlet (very high), evaporator outlet
Subcooling	Temperature below saturation, in liquid form	Condenser outlet (~8-12°F typical)

Formulas

> ***Superheat = Actual Vapor Temperature − Saturation Temperature (at that pressure)***
>
> ***Subcooling = Saturation Temperature − Actual Liquid Temperature***

Diagnostic interpretation (TXV-controlled system)

Subcooling	Indicates
Low (<5°F)	Undercharge OR restriction before condenser
Normal (8-12°F)	Charge is correct
High (>15°F)	Overcharge OR restriction after condenser

[redacted]

CFC/HCFC molecules:

1. Released at ground level
2. Don't dissolve in water → don't rain out
3. Chemically stable in troposphere (no UV-C, no hydroxyl-reactive H if CFC)
4. Drift up to stratosphere over years/decades
5. [redacted]
6. Cl atom catalytically destroys ~100,000 ozone molecules:
    - [redacted]
    - $ClO + O \rightarrow Cl + O_2$ (Cl regenerated)
7. Cl eventually removed by side reactions

Key facts:

- "Non-condensable" gases like CO_2 ≠ ozone depletion
- Only **chlorine and bromine** destroy ozone
- HFCs and HFOs have **no chlorine** → ZERO ozone impact
- Fluorine atoms released from HFCs form HF (stable, no ozone reaction)

[redacted]

Certification	What you can work on
Type I	Small appliances (≤5 lbs refrigerant) — fridges, freezers, water coolers, dehumidifiers, vending machines
Type II	**High-pressure** systems — most residential AC, commercial refrigeration
Type III	**Low-pressure** systems — centrifugal chillers (R-11, R-123)

Universal	All three above (Core + I + II + III in one cert)

Important EPA terminology:

- Type III = **"low-pressure"** (not "vacuum pressure," not "negative pressure")
- [redacted]
- "Non-condensable" = **Air contamination** (not moisture)

HC Refrigerants Approved for Domestic Small Appliances

Refrigerant	Application	Charge limit
R-600a (isobutane)	New household refrigerators, freezers, dehumidifiers	≤ 57 grams
R-290 (propane)	Some commercial self-contained refrigeration	Charge-limited per ASHRAE 15
R-441A	Some retrofit applications	Application-specific

PART 3 — CONCEPT EXPLANATIONS (deeper [redacted]

Why low-pressure systems behave differently

Low-pressure refrigerants (R-11, R-123) have **boiling points near room temperature at atmospheric pressure** (~75°F and ~82°F respectively). This creates several unique characteristics:

1. **Normal operation runs in vacuum** — evaporator pressure is below atmospheric
2. **Air leaks INWARD** (not outward) — opposite of high-pressure systems
3. **Purge units required** — to continuously remove air that has leaked in
4. [redacted]
5. **Deeper recovery vacuum required** — 25 mm Hg absolute (nearly perfect vacuum) because the refrigerant is so close to atmospheric pressure
6. [redacted]

[redacted]

The EPA scales recovery requirements with system size because:

- More refrigerant = more environmental impact if vented

- Larger systems are typically newer and better-equipped
- Larger commercial systems have more technical resources
- Above 200 lbs, the EPA requires deeper recovery vacuum

The 50-lb threshold for leak repair is a related concept — at 50+ lbs, the system is large enough that leak [redacted]

Why ASHRAE class A1 dominates older refrigerants

Old refrigerants (R-12, R-22, R-134a, R-410A) were designed before environmental concerns — they were optimized for:

[redacted]

- Non-flammability

[redacted] makes them safe to breathe makes them persist for decades in the atmosphere — high GWP and (for CFCs/HCFCs) high ODP.

Modern low-GWP refrigerants accept some flammability (A2L) or higher toxicity (R-123's B1) in exchange for environmental benefits. The industry trade-off has shifted.

[redacted]

[redacted] exposure has documented health impacts.

Because of B1 classification, R-123 chillers require:

- Refrigerant monitors in the mechanical room
- Enhanced ventilation
- More rigorous PPE during service
- Higher reporting requirements for leaks

Why HFOs are the future

HFOs have:

- **No chlorine** → zero ODP
- [redacted]
- **C=C double bond** → extremely reactive in lower atmosphere → very short atmospheric life (days to weeks)
- [redacted]

R-1234yf (auto AC), R-1234ze (chillers, refrigeration), R-1233zd (chillers replacing R-123) are the three major commercial HFOs.

Why "non-condensable" terminology is confusing

The HVAC industry uses "non-condensable" specifically to mean **air contamination** — gases like nitrogen and oxygen that cannot be condensed back to liquid under refrigeration system conditions. Water vapor isn't called a non-condensable even though it can condense at HVAC temperatures, because it's handled as a [redacted]

When you see "non-condensable" on the exam, translate it in your head to **"air."**

Why "vacuum pressure" isn't an EPA term

[redacted] EPA's official terminology when answering exam questions.

[redacted]

The Rule of 90 lets you decode any refrigerant R-number into its molecular formula:

> ***R-number + 90 = (C)(H)(F) as digits***
>
> ***Cl = (2·C + 2) – H – F***

[redacted]

- Decoding unfamiliar refrigerants
- [redacted]
- Calculating composition for chemistry problems

For HFOs (R-1000+), the rule still works after subtracting the leading 1 (e.g., R-1234yf → 234, then 234+90 = 324). The "1" prefix indicates the C=C double bond.

QUICK-REFERENCE CHEAT SHEET (last-minute review)

Print this section and review the morning of the exam.

[redacted]

Number	What
4" Hg	Type I recovery vacuum
5 lbs	Small appliance / recordkeeping threshold

10 psig	Low-pressure leak test MAX
10" Hg	High-pressure ≥200 lb recovery / Medium-pressure <200 lb recovery
15 psig	Low-pressure rupture disc burst
15" Hg	Medium-pressure ≥200 lb recovery
15 lbs	System-dependent recovery cutoff
20%/year	Comfort cooling leak threshold
25 mm Hg absolute	Low-pressure recovery vacuum
30%/year	Commercial/industrial leak threshold
30 days	Repair OR plan deadline
50 lbs	Leak repair rule activation
50 ppm	R-123 OEL
57 grams	R-600a charge limit (household)
90%	Type I operating-compressor recovery
80%	Type I non-operating-compressor recovery
200 lbs	Charge size threshold for recovery vacuum tier
12 months	Standard retrofit/retire deadline
18 months	Extended (50+ lbs exempt) retrofit/retire
Nov 15, 1993	Recovery equipment certification cutoff date

Refrigerant identifications

Refrigerant	Class	Family	Pressure	Notable
R-11	A1	CFC	Low	Banned, chiller refrigerant
R-12	A1	CFC	Medium	Banned, auto AC
R-22	A1	HCFC	High	Phased out (production), still in service
R-32	A2L	HFC	High	Modern low-GWP, slightly flammable
R-123	B1	HCFC	Low	Modern chiller, toxic
R-134a	A1	HFC	High	Auto AC, chillers

R-290	A3	HC	n/a	Propane, highly flammable
R-410A	A1	HFC blend	High	Most residential AC
R-454B	A2L	HFC/HFO blend	High	R-410A replacement
R-600a	A3	HC	n/a	Isobutane, household fridges
R-717	B2L	Natural	n/a	Ammonia, industrial
R-744	A1	Natural	Very high	CO_2, transcritical
R-1234yf	A2L	HFO	High	Auto AC, low GWP
R-1233zd	A1	HFO	Low	R-123 replacement

Translation table for exam jargon

Exam term	Plain English
Non-condensable	Air contamination
HC	Hydrocarbon (NOT HCFC)
System-dependent	Passive (no internal pump)
Self-contained	Active (has internal pump)
Saturation temperature	Temp where phase change happens at a pressure
Superheat	Vapor temperature above saturation
Subcooling	Liquid temperature below saturation
Low-pressure (Type III)	Centrifugal chillers (R-11, R-123)
High-pressure (Type II)	Most residential/commercial AC
Small appliance (Type I)	≤5 lbs refrigerant
Exempt refrigerant	Non-ODS (HFC, HFO) — not subject to ODS phaseout
Reclamation	Returning refrigerant to virgin spec (AHRI 700)
Recycling	On-site cleanup, less rigorous than reclamation

Top 10 exam patterns to recognize

1. **"Maximum pressure for leak testing low-pressure system"** → 10 psig (NOT 15 — that's the rupture disc)

2. **"Recovery vacuum for low-pressure system"** → 25 mm Hg ABSOLUTE
3. **"Self-contained recovery equipment definition"** → has its own internal pump
4. **"System-dependent recovery limit"** → 15 lbs or less
5. **"Type I operating compressor recovery"** → 90% or 4" Hg vacuum
6. **"Non-condensable example"** → AIR (not moisture)
7. [redacted]
8. **"Refrigerants R-12, R-22, R-134a are classified as"** → A1
9. **"Strong odor during recovery"** → compressor burn-out
10. **"Records required for 5-50 lb appliances"** → monthly records of added/recovered/reclaimed (by type)

[redacted]

Final exam strategy

1. **Read every question twice** before picking. Most misses happen on first-pass reading errors.
2. **Identify the certification type** (I, II, III) before applying any rule — the same word means different [redacted]
3. **Default to EPA's exact terminology** — don't pick clever-sounding answers if they use non-EPA words.
4. **15 is rarely the right answer.** When it appears, double-check. Most often it's a distractor.
5. **For recovery vacuum questions**, identify (a) refrigerant pressure category, (b) charge size, (c) equipment age — then look up the right cell in the table.
6. **For ASHRAE classification questions**, default to A1 unless the refrigerant is specifically R-32, [redacted]
7. **Don't run out of time on Core.** Core questions are the hardest. Type sections are easier. Pace yourself.

Study guide compiled June 1, 2026. Based on ESCO 9th Edition Manual practice exam results + Universal certification scope (Core + Type I + Type II + Type III).

[redacted]